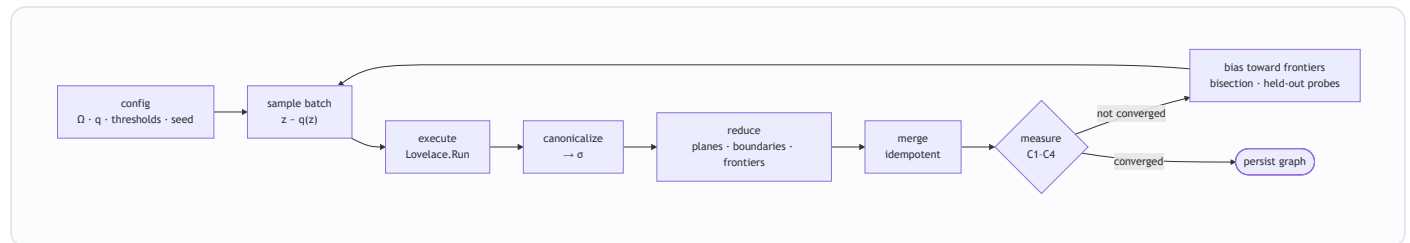


The learned behavior graph — visual companion

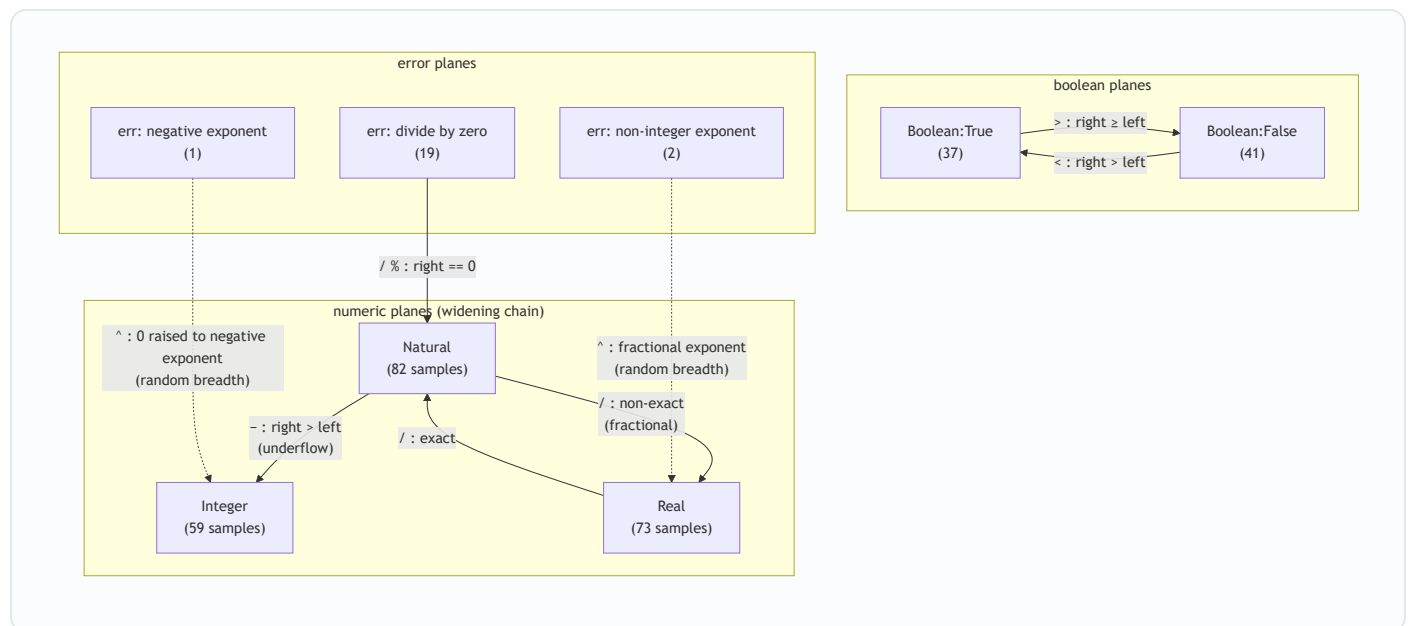
Mermaid diagrams for the graph discovered by Lovelace.Knowledge.Run against the real Lovelace.Run engine (seed 20240617, 314 samples, converged C1-C4). See CONVERGENCE-RESULTS.md for the numeric evidence.

1. The discovery pipeline



2. The behavior planes and their boundaries

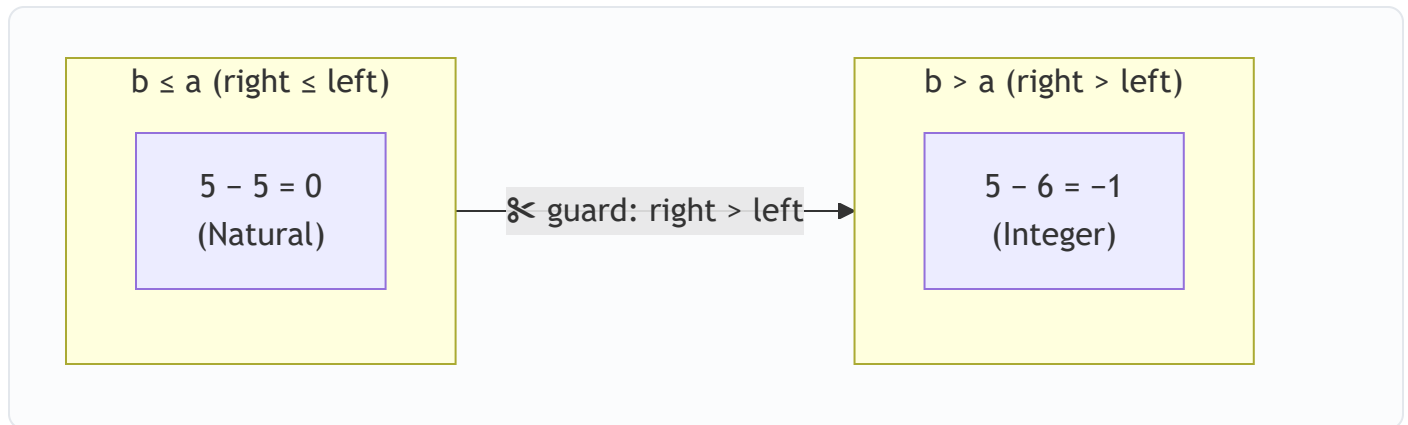
Eight planes were found purely from observations, grouped into three families. Edges are the localized boundaries (each carries a fitted guard and counterexamples on both sides).



The dashed edges mark planes observed through random sampling that have no localized guard — honest open-world findings, not forced connections.

3. Natural-subtraction underflow (a localized threshold boundary)

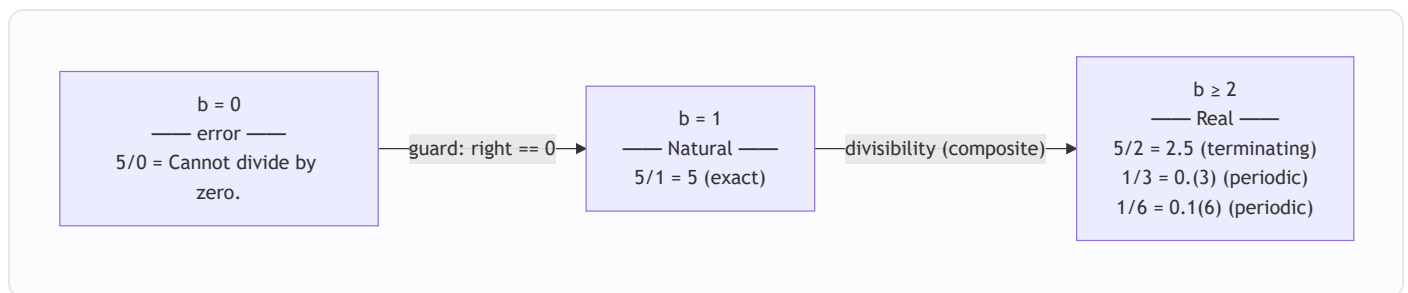
Sweeping the right operand b over naturals with a fixed left a , the result stays Natural while $b \leq a$ and flips to Integer the moment $b > a$. The reducer bisects the changed interval down to the exact integer and fits the guard "right > left".



The same boundary re-appears at every anchor — $0-0/0-1$, $1-1/1-2$, $2-2/2-3$, $5-5/5-6$, $10-10/10-11$ — so the guard "right > left" is learned five times and is stable across two step sizes ($h=2$, $h=3$), which is exactly the C2 check.

4. Division: one operation, three adjacent planes

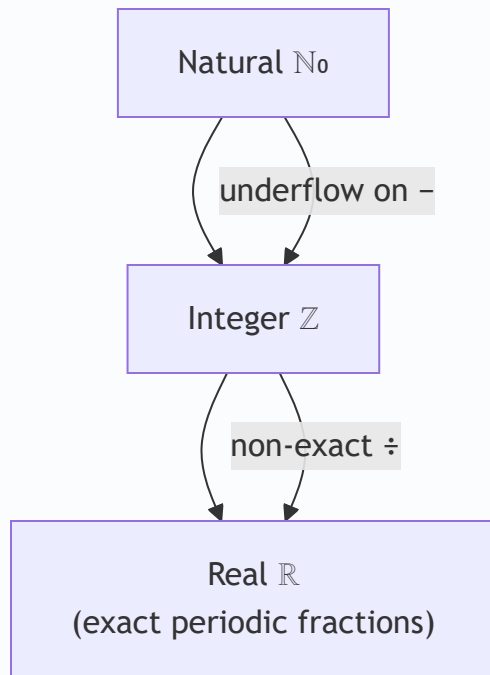
Division over naturals exposes a three-way split along the divisor b :



The Natural $\rightarrow$ Real edge is not a clean interval (it flips whenever the divisor stops dividing the dividend), so the reducer marks it composite and reports the bounding samples instead of inventing a predicate — the open-world rule.

5. The widening lattice the numeric planes reveal

The numeric planes are ordered by the engine's widening rule; this lattice is inferred from the observed result kinds, not read from source:



6. Where the 314 samples went

"Sample support per behavior plane (314 samples)"

